

1 Environment



Water-pipe digital twin
Controlled disturbances



Human Machine Material Logistics

Machines, workers, buffers,
gantries, and transport robots

One disturbance axis at a time
so bottleneck traces stay attributable

2 Data processing

Minute windows

Raw logs become a
resource--time grid

Process-informed labels

Type-specific rules plus
persistence filtering

Short-pattern clustering

Recurrent precursors,
not the bottleneck label



process · buffer · logistics

3 BNPDFormer

Grouped resource embedding

queue · stall · logistics · type

Temporal

one resource
over time

Graph

neighbours
+ patterns

Semantic

similar, not
only adjacent

Pattern keys condition geographic attention
so delayed propagation can be learned

Prediction heads

Station event · process cause

Remaining order time

4 Prediction report

Station

Which resource constrains
throughput

Onset and duration

When it starts and
how long it persists

Process cause

Shortage, transport delay,
starvation, or queue buildup

Remaining order time

Line-level consequence
of the current state